

Models and methods

DS_05 | Social media sentiment and misinformation analysis | October 2026

We analyse 19,787 public Reddit posts and YouTube comments about Australia's cost of living, 2019-2025. Tone, topic and factual credibility are different outputs. Neither a negative sentiment label nor a wording flag establishes that a claim is false.

Task and method	Actual input and output	Why used and important limits
Sentiment: adapted RoBERTa Base: cardiffnlp/twitter-roberta-base-sentiment-latest	Author text -> positive, neutral or negative. Complete text in 512-token windows, combined by token count. Class scores and mixed windows flag readings for review.	Locally fine-tuned classifier and final two blocks on 112 training posts. Square-root class weighting; 62 development examples select epoch 1. Sarcasm, headlines and rare classes remain difficult.
Baseline: VADER Lexicon and rules	Original compound score; default cutoffs +/-0.05 retained for comparison. Domain labels: positive at least +0.995, negative at most -0.05; otherwise neutral. Emotional-rule intensity is unchanged.	Fast and inspectable. Handles punctuation/capitals and social vocabulary; weak on context. Strong emotion requires both intensity and shouting before a rule fires.
Topics: BERTopic all-MiniLM-L6-v2 + UMAP + HDBSCAN + c-TF-IDF	Text embeddings -> discussion clusters and characteristic keywords. UMAP seed 42; HDBSCAN minimum cluster 60; ten displayed topics.	Groups similar discussions without truth labels. Human naming/interpretability review is still required. TF-IDF + NMF is a recorded fallback, not silently described as BERTopic.
Misinformation review queue Five explicit-source-cue rules v3	Author text, preserved source cues and VADER intensity -> matched evidence, weighted score and review priority.	A transparent prototype where domain veracity training labels are not available. Weights and thresholds are hand-chosen, uncalibrated assumptions. It is not a trained fake/real classifier.
Worked claim checks Dated source comparison	Selected factual claim + original context + primary evidence -> limited finding: misleading, supported, satire or unverified.	Shows why a specific claim is wrong or supported. Six purposive examples; research review is AI-assisted and team sign-off is pending. This is not full-corpus fact checking.

Validation status

Measured sentiment comparison: On the same 36-post internal test subset: base RoBERTa 77.78% accuracy (macro F1 0.540); adapted RoBERTa 83.33% (0.562); default VADER 36.11% (0.327); calibrated VADER 63.89% (0.428). Training, development and test IDs are disjoint; overlapping videos are excluded from training. The 85% target remains unmet. This small retrospective comparison is not population-wide accuracy.

Misinformation: 1,357 medium/high-priority candidates out of 19,787 posts. This describes the review queue, not confirmed misinformation prevalence. The one worked misleading finding is a specific arithmetic error, not a verdict on the author or every sentence.

Sources: [RoBERTa model card](#); [VADER](#); [BERTopic documentation](#).

Why these five rules and how scores work

The five cues cover certainty, numerical evidence, emotion, presentation and attribution. They were selected because each is inspectable and cheap to reproduce. Research supports examining language/context cues, but does not validate these exact regexes, weights or thresholds.

Rule and weight	What the code detects	Why inspect it and the next check
Absolute certainty 0.28	Listed phrases such as always, never, everyone knows and guaranteed.	Categorical claims may exceed their evidence. Check population and exceptions. A universal statement can still be true.
No recorded source cue for numbers 0.24	Percentages or scaled quantities without a URL or recognised institution name. Generic study, report or according to phrases do not identify a source.	Check source, date and denominator. An uncaptured source may exist; a source mention alone does not prove correctness.
Strong emotion + shouting 0.20	Absolute VADER compound above 0.60 with capital-letter ratio above 18%, or above 0.85 with capitals above 10%; at least 20 letters.	Separate the factual assertion from emotional delivery. Anger about a genuine problem can trigger the rule.
Sensational presentation 0.14	Repeated ! or ?, or listed words such as shocking and outrageous.	Read beyond the framing and compare the claim with evidence. Satire, frustration and accurate breaking news can match.
Hard-to-trace attribution 0.14	Phrases such as sources say, someone told me, apparently and allegedly.	Locate the source and assess its relevance. Responsible reporting can use qualified attribution.

Calculation and evidence example

Post score: add only the weights of matched rules (maximum 1). Below 0.30 = low priority; 0.30 to below 0.55 = medium; at least 0.55 = high. Rule counts overlap because one post can match several rules. No score is a probability of falsity.

Keyword association index: (candidate posts with word / all posts with word) / overall candidate rate, multiplied by $\log(1 + \text{candidate count}) / \log(1 + \text{maximum candidate-word count})$. A word is counted once per post. This index has a different scale from the post priority score and can exceed 1.

Worked check: a post adds 7.8% (2022 CPI) and 4.1% (2023 CPI), claiming 11.9% in two years. The compounded change from those rounded annual rates is $(1.078 \times 1.041 - 1) \times 100 = 12.2198\%$, approximately 12.2%. The error concerns this calculation; an individual household's experience can differ from CPI.

Evidence: [ABS December 2022 CPI](#) and [ABS December 2023 CPI](#).

Research and interpretation

Language research: [Rashkin et al. 2017](#); [Volkova et al. 2017](#). Emotion/belief: [Martel et al. 2020](#) concerns readers, not a validated emotional-post detector. Evidence boundary: [Fact-checking or gut-checking? 2022](#).

Interpreting a finding: point to the exact claim, dated evidence and limited finding. If only rules matched, call it a candidate for review. Never infer truth from sentiment, a strong colour, a keyword or a high score. Rule precision/recall and corpus misinformation prevalence remain unverified.